### 2.4.4 Conversion of fractional Decimal to Binary

The method of **repeated multiplication by 2** has to be used to convert such kind of decimal fractions.

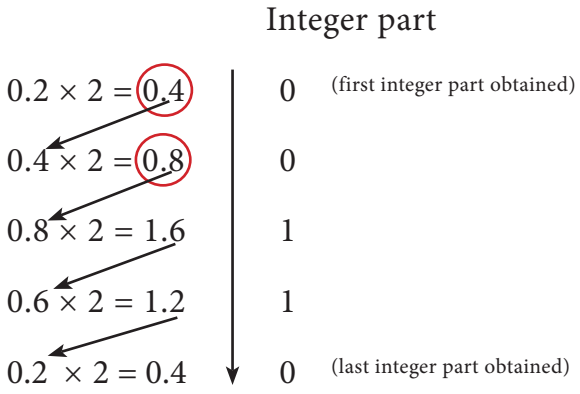
The steps involved in the method of **repeated multiplication by 2**:

Step 1: Multiply the decimal fraction by 2 and note the integer part. The integer part is either 0 or 1.

Step 2: Discard the integer part of the previous product. Multiply the fractional part of the previous product by 2. Repeat Step 1 until the same fraction repeats or terminates (0).

Step 3: The resulting integer part forms a sequence of 0s and 1s that become the binary equivalent of decimal fraction.

Step 4: The final answer is to be written from first integer part obtained till the last integer part obtained.




**Note:** Fraction repeats, the product is the same as in the first step.

Write the integer parts from top to bottom to obtain the equivalent fractional binary number. Hence $(0.2)_{10}=(0.00110011\ldots)_2 = (0.00110011)_2$

**Workshop**

3. Convert the following Decimal numbers to its equivalent Binary, Octal, Hexadecimal.

1) 1920 2) 255 3)126

### 2.4.5 Binary to Decimal Conversion

To convert Binary to Decimal we can use positional notation method.

Step 1: Write down the Binary digits and list the powers of 2 from right to left(Positional Notation)

Step 2: For each positional notation written for the digit, now write the equivalent weight.

Step 3: Multiply each digit with its corresponding weight

Step 4: Add all the values.

*Table 2.3 Positional Notation and Weight*

| Positional Notation | Weight | Positional Notation | Weight |
|---|---|---|---|
| $2^0$ | 1 | $2^6$ | 64 |
| $2^1$ | 2 | $2^7$ | 128 |
| $2^2$ | 4 | $2^8$ | 256 |
| $2^3$ | 8 | $2^9$ | 512 |
| $2^4$ | 16 | $2^{10}$ | 1024 |
| $2^5$ | 32 | | |

**Example**

Convert $(111011)_2$ into its equivalent decimal number.

| Weight | 32 | 16 | 8 | 4 | 2 | 1 |
|---|---|---|---|---|---|---|
| Positional Notation | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
| Given number | 1 | 1 | 1 | 0 | 1 | 1 |